# PhonePe Transaction Insights

This analytical report presents a complete data exploration and business interpretation of PhonePe transaction data. The project focuses on transaction amount, transaction volume, state-level performance, district-level trends, category-wise contribution, and seasonal behavior across time. The goal is to convert raw transactional records into actionable business intelligence for strategic decisions.

## 1. Project Problem Statement

With the increasing adoption of digital payment systems, it becomes essential to understand user payment behavior, geographical transaction patterns, category-wise contribution, and overall platform growth. The purpose of this project is to analyze PhonePe transaction data, identify top-performing regions and categories, detect time-based trends, and derive business insights that can support decision-making, targeting, and product strategy.

The project is based on the broader objective of analyzing transaction dynamics, user engagement, geographical insights, payment performance, trend analysis, and strategic business recommendations through SQL, Python, and dashboarding workflows.

## 2. Project Summary

This project performs structured exploratory data analysis on PhonePe transactional records. The analysis includes 15 visualizations divided into univariate, bivariate, and multivariate sections, along with 3 statistical hypothesis tests. Each visual has been interpreted from both analytical and business perspectives. The final outcome is a report that explains market concentration, growth patterns, category dominance, quarter-wise variation, and transaction value behavior.

The analytical workflow covers data cleaning, feature engineering, descriptive analysis, visual exploration, statistical testing, and synthesis of findings into client-ready recommendations.

# 3. Exploratory Data Analysis

The EDA is divided into three levels: univariate analysis for understanding individual variables, bivariate analysis for studying relationships between two variables, and multivariate analysis for uncovering complex interaction patterns.

## Univariate Analysis

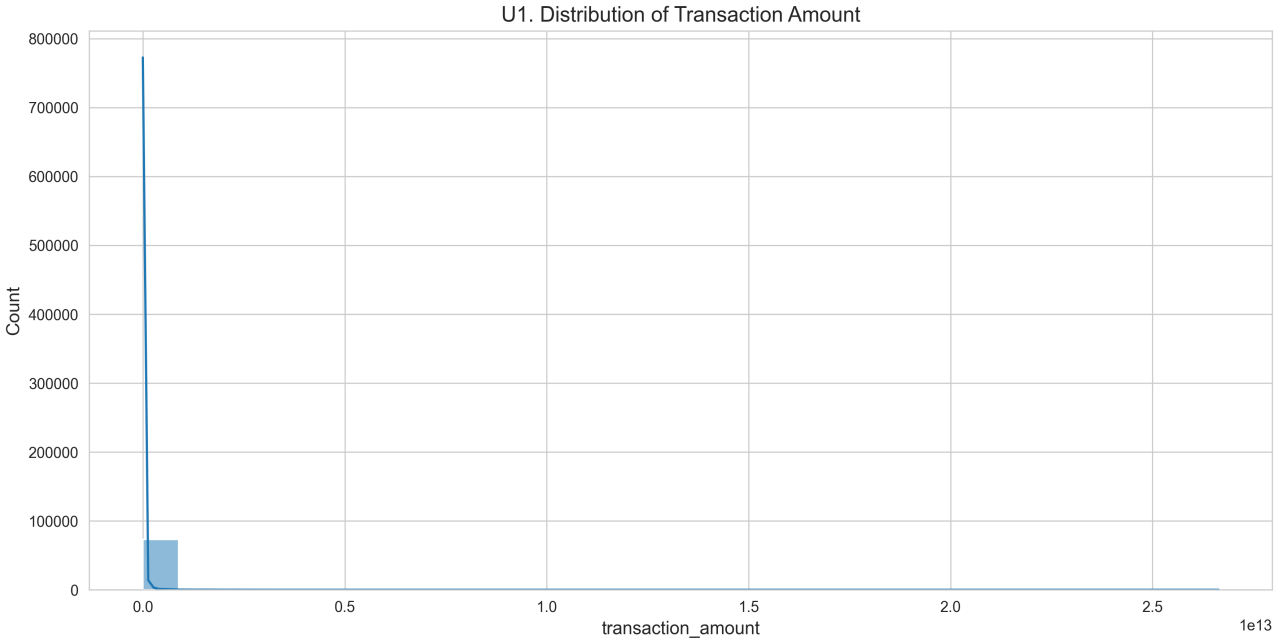
### U1. Distribution of Transaction Amount

**Why this chart was chosen:** A histogram is chosen to understand the spread, skewness, concentration, and possible outliers in transaction amount.

**Results:** Transaction amount ranges from 31.00 to 26,633,866,936,498.00. Mean amount = 24,390,808,051.95, Median = 173,754,817.48.

**Insights:** The distribution helps identify whether transaction values are concentrated in a narrow band or spread widely. A strong right skew usually indicates that a small number of records contribute very large transaction values compared with the majority.

**Result:** Transaction amount distribution reveals whether the platform is driven by many small transactions or a mix including very large high-value activity.



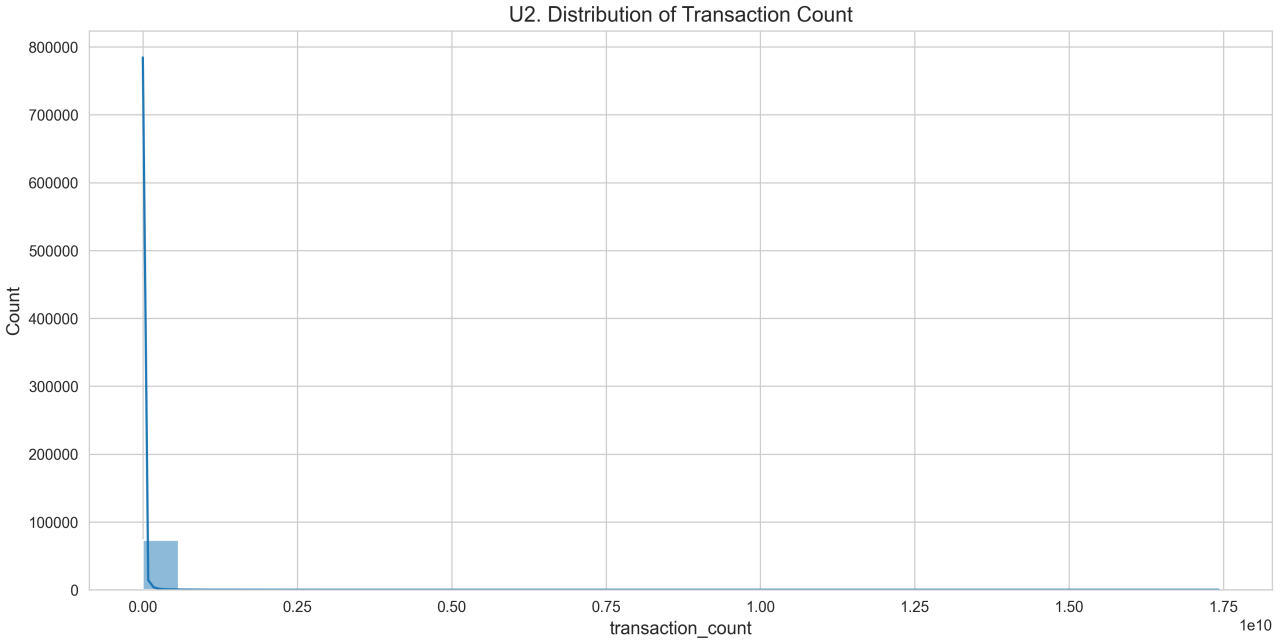

### U2. Distribution of Transaction Count

**Why this chart was chosen:** A histogram is used to inspect the frequency distribution and concentration of transaction volume.

**Results:** Transaction count ranges from 1 to 17,419,191,808. Mean count = 16,835,842.32, Median = 102,683.00.

**Insights:** This chart shows whether user activity volume is balanced or concentrated. If highly skewed, a few state/district/category combinations may be dominating overall transaction traffic.

**Result:** Transaction count distribution shows the operational load pattern and identifies heavy activity concentration.



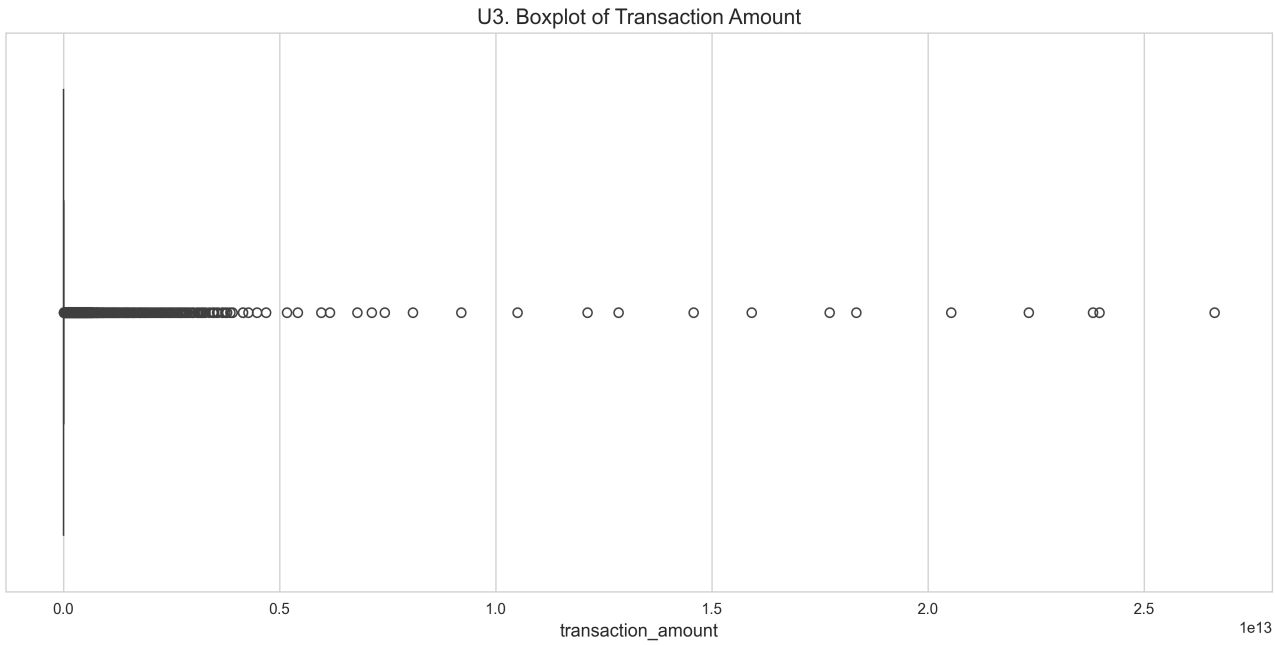

## U3. Boxplot of Transaction Amount

**Why this chart was chosen:** A boxplot is chosen to quickly detect spread, median, quartiles, and extreme values in transaction amount.

**Results:** Q1 = 857,406.00, Q3 = 4,538,170,194.83, IQR = 4,537,312,788.83, Estimated high-end outliers = 12522.

**Insights:** The boxplot highlights unusually large transaction amount values. These may represent peak market regions, seasonal effects, or exceptional transaction clusters that deserve separate business attention.

**Result:** Outlier analysis helps identify exceptional payment hotspots and abnormal transaction behavior.



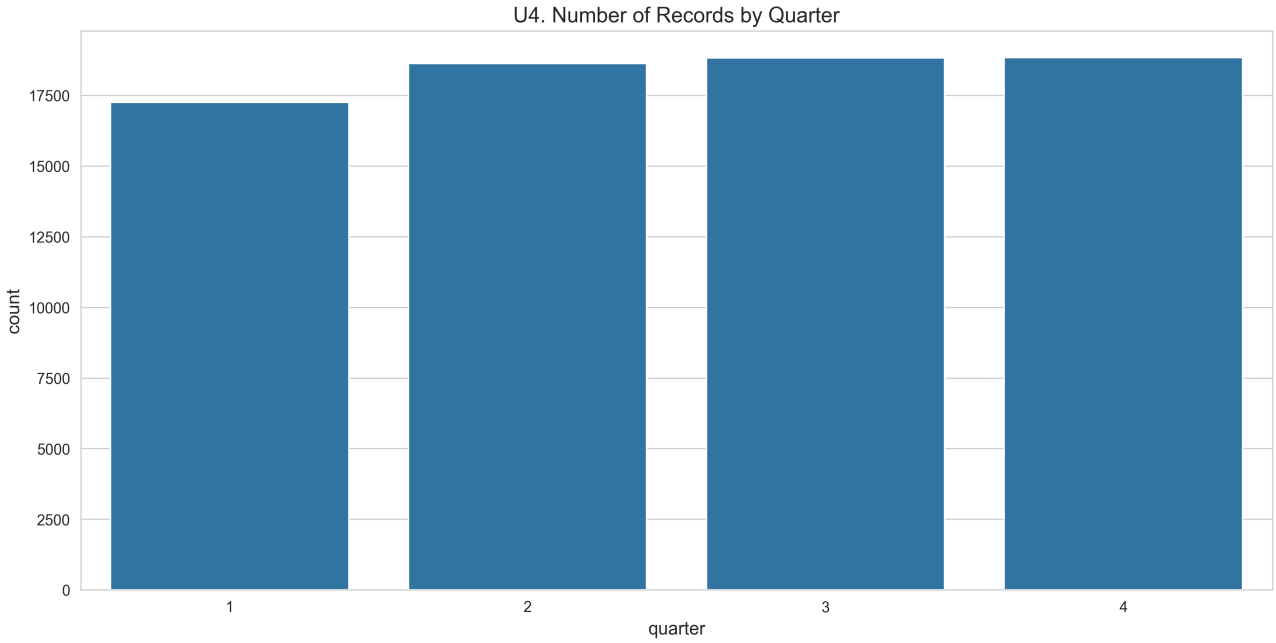

## U4. Number of Records by Quarter

**Why this chart was chosen:** A countplot is used to verify quarter-level data coverage and temporal representation.

**Results:** Quarter-wise record distribution: {1: 17249, 2: 18629, 3: 18828, 4: 18834}

**Insights:** This chart checks whether records are evenly represented across quarters. Uneven distribution may indicate data incompleteness or stronger reporting concentration in particular quarters.

**Result:** Quarter distribution helps validate data completeness and seasonal representation.



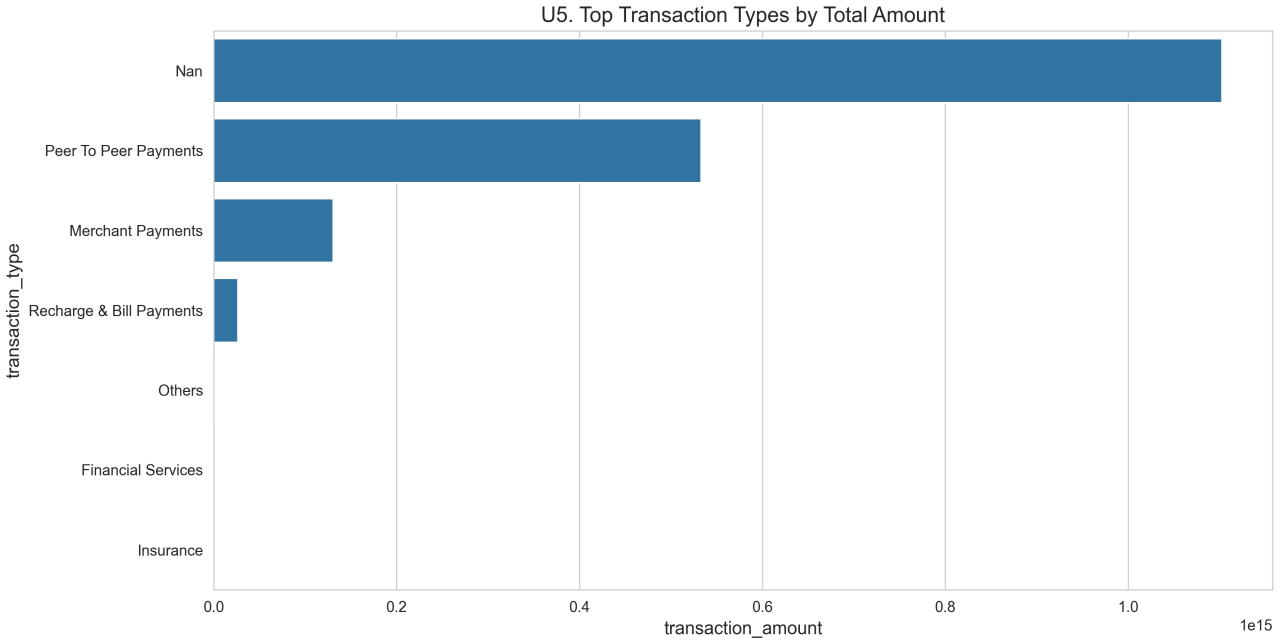

## U5. Top Transaction Types by Total Amount

**Why this chart was chosen:** A horizontal bar chart is chosen for easy comparison of total transaction value across payment categories.

**Results:** Top transaction types by total amount:
transaction_type transaction_amount
Nan 1.102614e+15
Peer To Peer Payments 5.330559e+14
Merchant Payments 1.306800e+14
Recharge & Bill Payments 2.667758e+13
Others 3.485623e+11
Financial Services 2.840397e+11
Insurance 4.003150e+10

**Insights:** This chart shows which payment categories contribute the highest monetary value. These are the most commercially important transaction categories for growth and optimization.

**Result:** Top payment categories reveal where most monetary throughput occurs.




## Bivariate Analysis

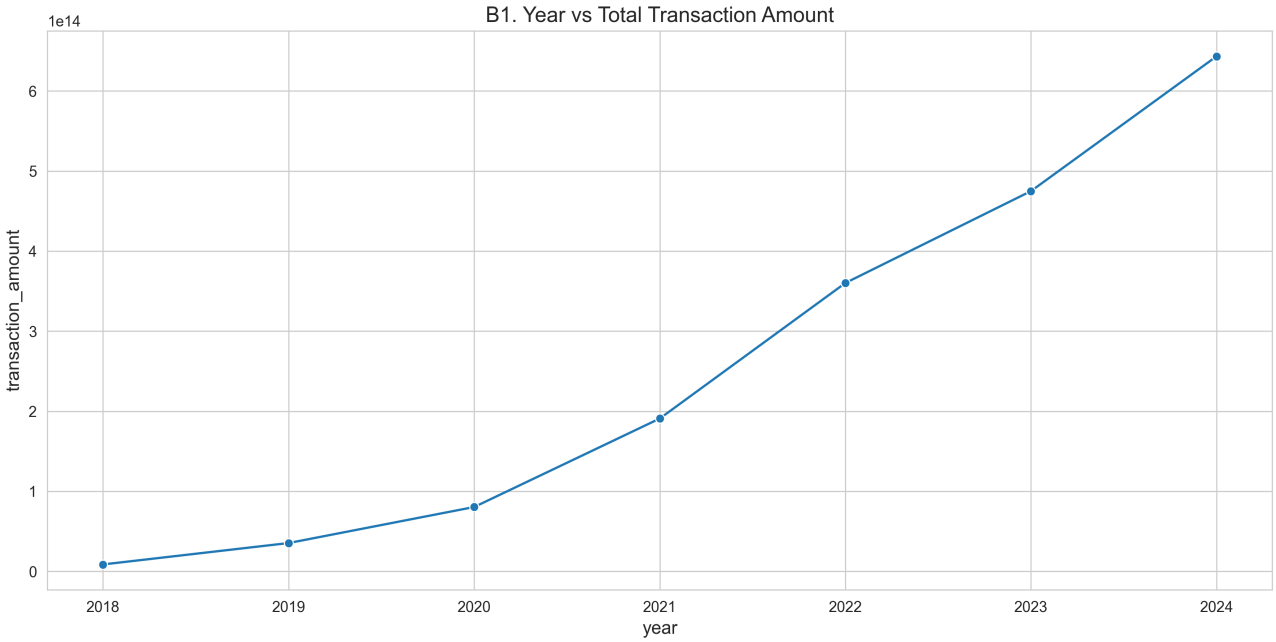
## B1. Year vs Total Transaction Amount

**Why this chart was chosen:** A line chart is best for analyzing changes over time and identifying long-term growth trends.

**Results:** Year-wise total transaction amount:
year transaction_amount
2018 8.694858e+12
2019 3.541450e+13
2020 8.041077e+13
2021 1.909175e+14
2022 3.601586e+14
2023 4.748796e+14
2024 6.432242e+14

**Insights:** This trend line helps detect yearly growth or decline in PhonePe transaction value. A rising trend reflects increasing platform adoption and higher digital payment dependence.

**Result:** Year-wise growth pattern shows whether transaction value is expanding steadily over time.



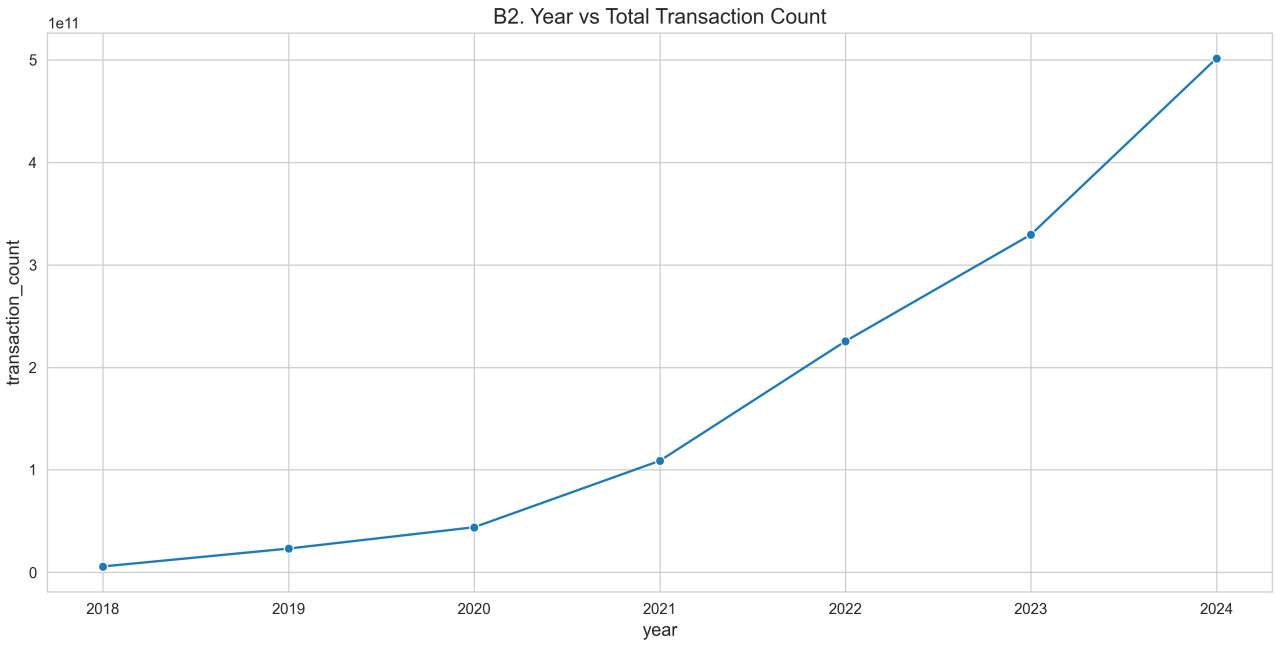

## B2. Year vs Total Transaction Count

**Why this chart was chosen:** A time-based line chart clearly shows whether transaction volume is increasing across years.

**Results:** Year-wise total transaction count:
year transaction_count
2018 5.764760e+09
2019 2.319951e+10
2020 4.401319e+10
2021 1.088114e+11
2022 2.254355e+11
2023 3.294808e+11
2024 5.014027e+11

**Insights:** This chart measures growth in usage volume. Comparing amount growth with count growth also helps determine whether growth is driven by more transactions or larger ticket sizes.

**Result:** Transaction count trend captures user activity growth and platform usage intensity.



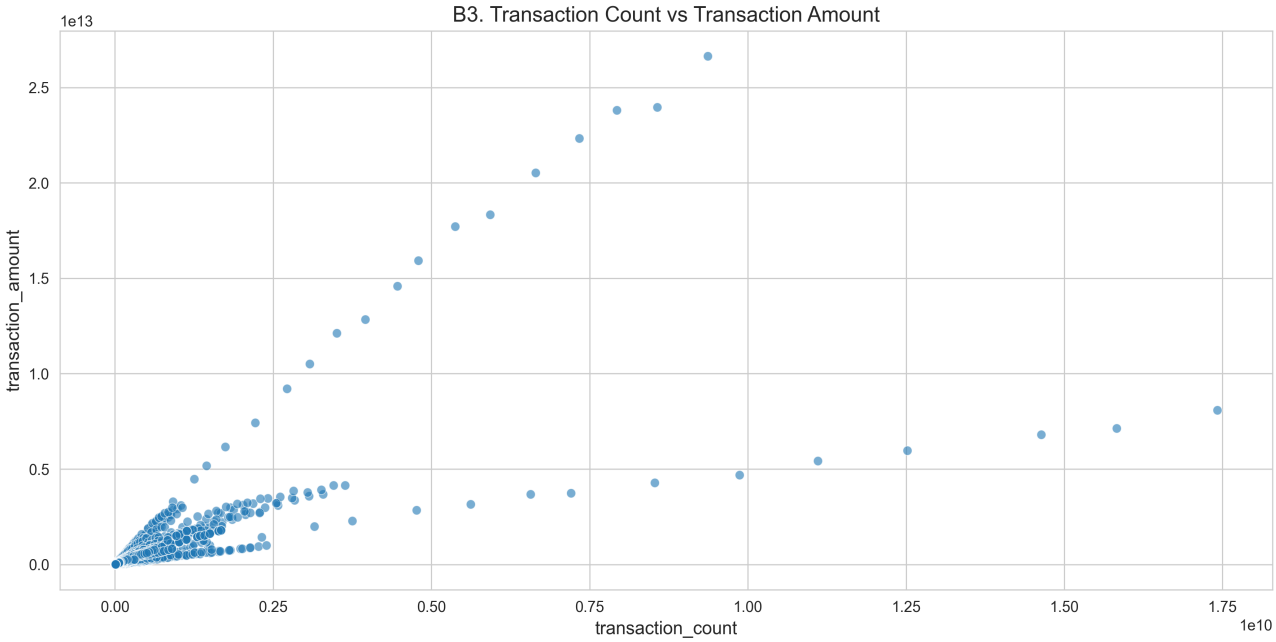
## B3. Transaction Count vs Transaction Amount

**Why this chart was chosen:** A scatter plot is chosen to examine the direct relationship between activity volume and monetary value.

**Results:** Pearson correlation between transaction count and amount = 0.7502, p-value = 0.000000

**Insights:** A strong positive correlation means locations or categories with higher number of transactions also tend to generate higher total value. Weak correlation would imply inconsistent ticket sizes or uneven usage behavior.

**Result:** Correlation analysis shows whether transaction amount scales proportionally with transaction volume.



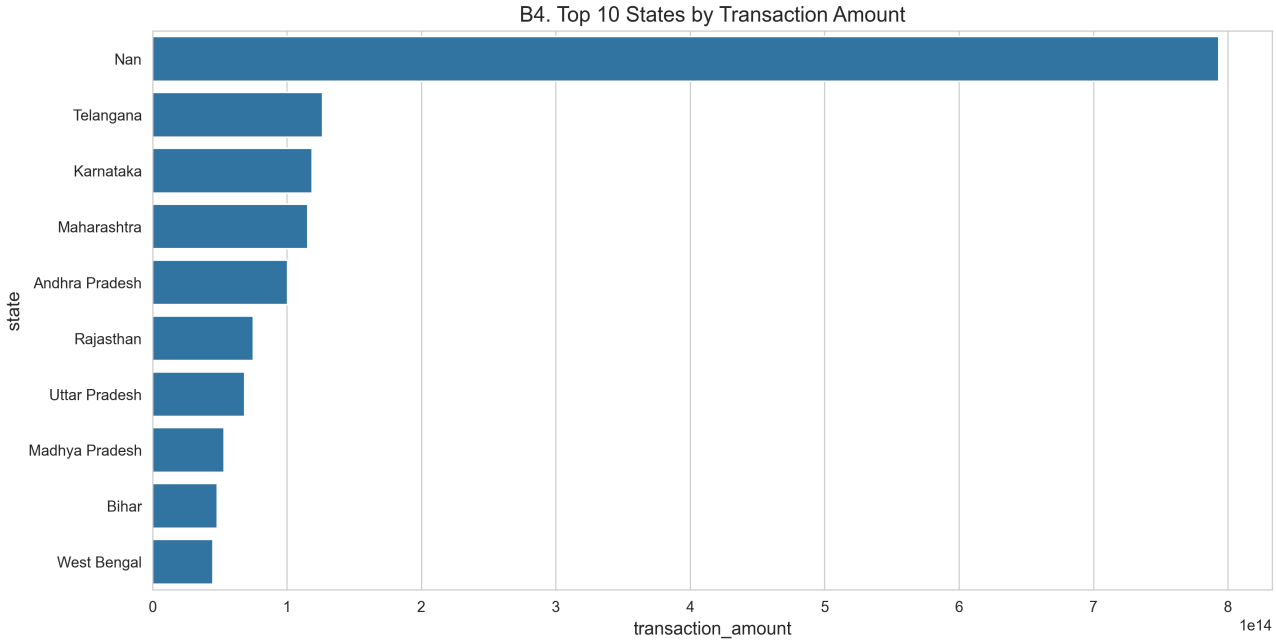

## B4. Top 10 States by Transaction Amount

**Why this chart was chosen:** A bar chart is ideal for ranking states by total transaction value.

**Results:** Top 10 states by total transaction amount:
state transaction_amount
Nan 7.933308e+14
Telangana 1.266604e+14
Karnataka 1.188653e+14
Maharashtra 1.155057e+14
Andhra Pradesh 1.006303e+14
Rajasthan 7.504683e+13
Uttar Pradesh 6.863485e+13
Madhya Pradesh 5.314716e+13
Bihar 4.809956e+13
West Bengal 4.502904e+13

**Insights:** This chart identifies the leading geographic contributors by value. These states are strong candidates for premium targeting, retention programs, and advanced service offerings.

**Result:** Top states reveal the strongest markets in terms of digital payment value.



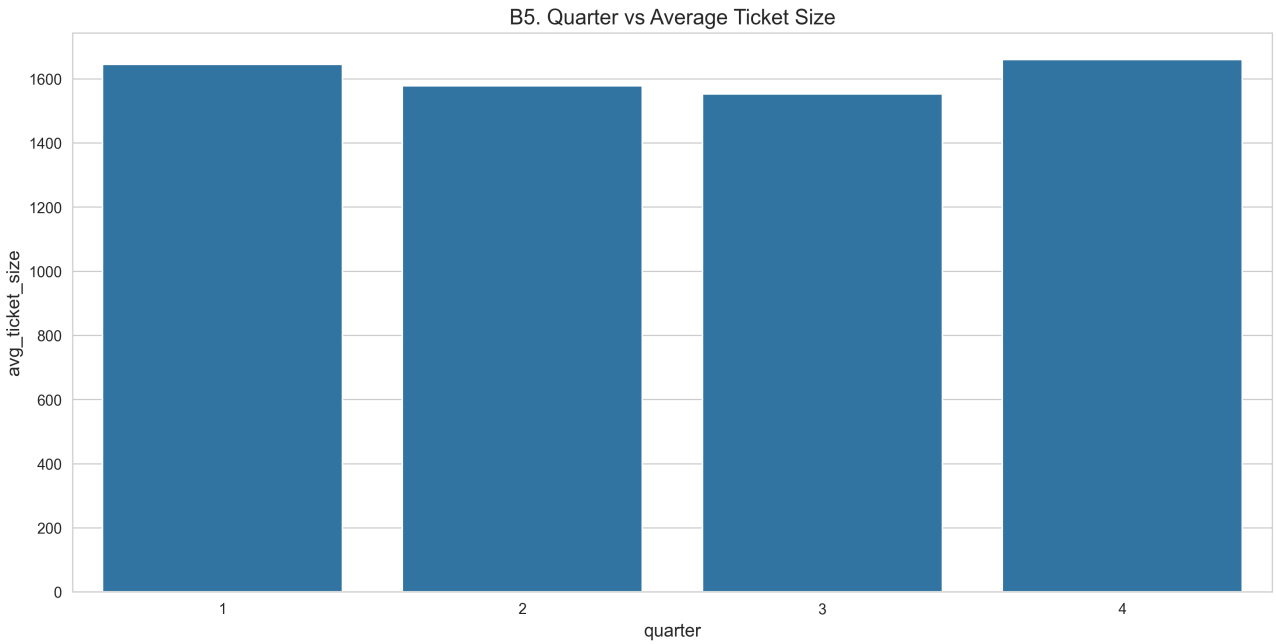

## B5. Quarter vs Average Ticket Size

**Why this chart was chosen:** A bar chart helps compare average transaction value across the four quarters.

**Results:** Quarter-wise average ticket size:
quarter avg_ticket_size
1 1644.748123
2 1577.782684
3 1552.196029
4 1659.518335

**Insights:** This shows whether customers spend more per transaction in specific quarters. It is useful for understanding seasonality, campaign timing, and high-value payment periods.

**Result:** Quarter-wise average ticket size helps detect spending seasonality and campaign opportunities.




## Multivariate Analysis

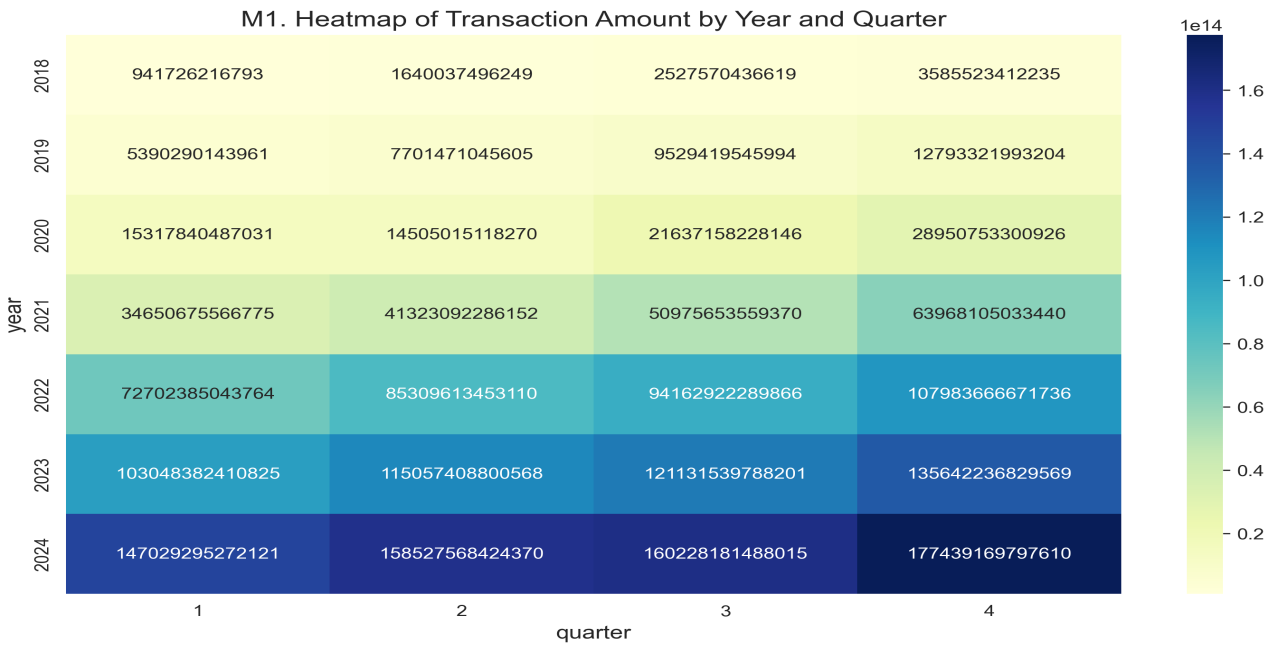
## M1. Heatmap of Transaction Amount by Year and Quarter

**Why this chart was chosen:** A heatmap is chosen to compare transaction amount across both year and quarter in a compact visual matrix.

**Results:** Year-Quarter transaction amount matrix:
quarter 1 2 3 4
year
2018 9.417262e+11 1.640037e+12 2.527570e+12 3.585523e+12
2019 5.390290e+12 7.701471e+12 9.529420e+12 1.279332e+13
2020 1.531784e+13 1.450502e+13 2.163716e+13 2.895075e+13
2021 3.465068e+13 4.132309e+13 5.097565e+13 6.396811e+13
2022 7.270239e+13 8.530961e+13 9.416292e+13 1.079837e+14
2023 1.030484e+14 1.150574e+14 1.211315e+14 1.356422e+14
2024 1.470293e+14 1.585276e+14 1.602282e+14 1.774392e+14

**Insights:** The heatmap reveals seasonal peaks, weak periods, and year-quarter combinations with the highest business volume. It is excellent for pattern spotting across two time dimensions simultaneously.

**Result:** Year-quarter heatmap exposes combined seasonality and long-term trend intensity.




## M2. Top States vs Transaction Type Heatmap

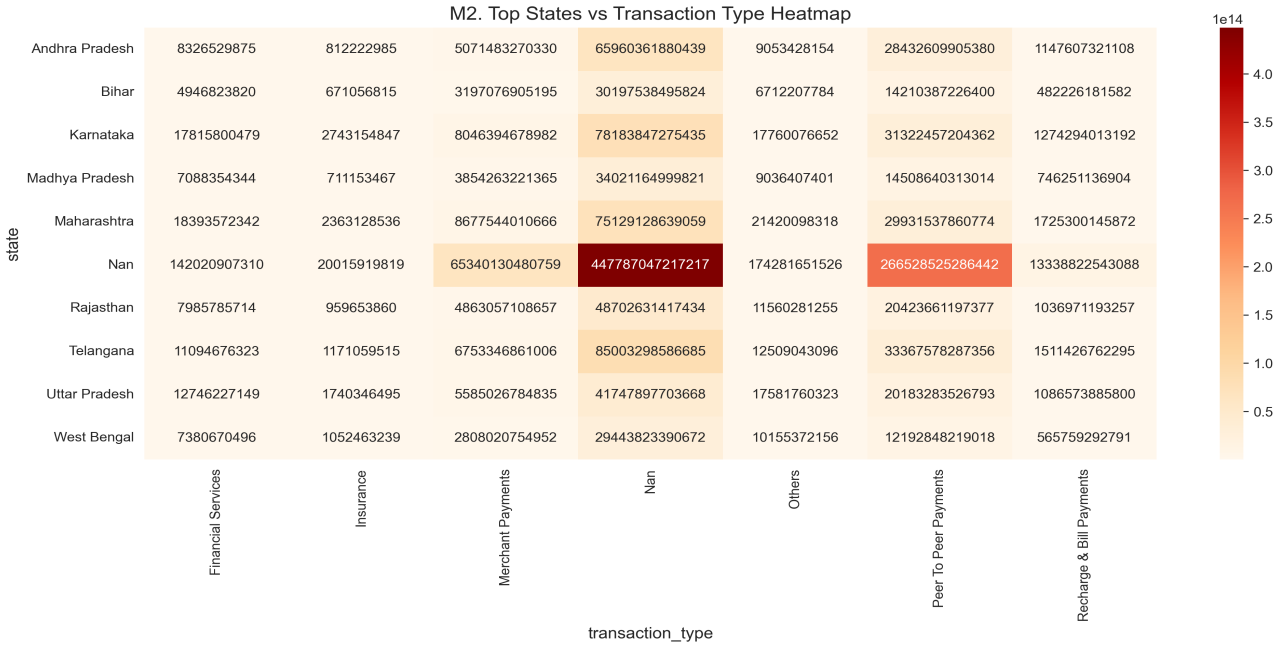
**Why this chart was chosen:** A heatmap is best for comparing many state-category intersections together.

**Results:** Top states by transaction type matrix:
transaction_type Financial Services Insurance Merchant Payments Nan Others Peer To Peer Payments Recharge & Bill Payments
state
Andhra Pradesh 8.326530e+09 8.122230e+08 5.071483e+12 6.596036e+13 9.053428e+09 2.843261e+13 1.147607e+12
Bihar 4.946824e+09 6.710568e+08 3.197077e+12 3.019754e+13 6.712208e+09 1.421039e+13 4.822262e+11
Karnataka 1.781580e+10 2.743155e+09 8.046395e+12 7.818385e+13 1.776008e+10 3.132246e+13 1.274294e+12
Madhya Pradesh 7.088354e+09 7.111535e+08 3.854263e+12 3.402116e+13 9.036407e+09 1.450864e+13 7.462511e+11
Maharashtra 1.839357e+10 2.363129e+09 8.677544e+12 7.512913e+13 2.142010e+10 2.993154e+13 1.725300e+12
Nan 1.420209e+11 2.001592e+10 6.534013e+13 4.477870e+14 1.742817e+11 2.665285e+14 1.333882e+13

Rajasthan 7.985786e+09 9.596539e+08 4.863057e+12 4.870263e+13 1.156028e+10 2.042366e+13 1.036971e+12
Telangana 1.109468e+10 1.171060e+09 6.753347e+12 8.500330e+13 1.250904e+10 3.336758e+13 1.511427e+12
Uttar Pradesh 1.274623e+10 1.740346e+09 5.585027e+12 4.174790e+13 1.758176e+10 2.018328e+13 1.086574e+12
West Bengal 7.380670e+09 1.052463e+09 2.808021e+12 2.944382e+13 1.015537e+10 1.219285e+13 5.657593e+11

**Insights:** This chart helps compare how transaction categories behave across leading states. It shows whether market leaders are driven by the same payment categories or different usage patterns.

**Result:** State-category comparison reveals regional preference differences in payment behavior.



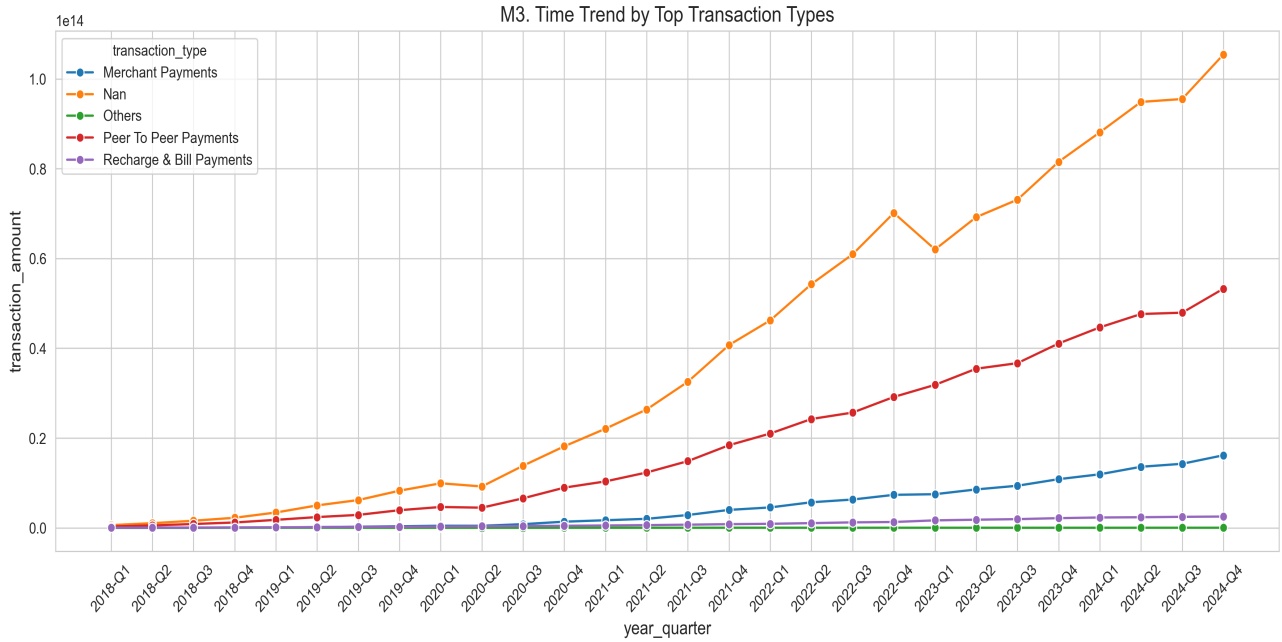

## M3. Time Trend by Top Transaction Types

**Why this chart was chosen:** A multi-line chart is chosen to compare category-level time trends across periods.

**Results:** Time trend by top transaction types prepared for top 5 categories.

**Insights:** This chart shows how leading transaction categories rise or decline over time. It is useful for identifying persistent winners, emerging categories, and volatile segments.

**Result:** Multi-category time trends reveal evolving payment behavior across leading transaction types.



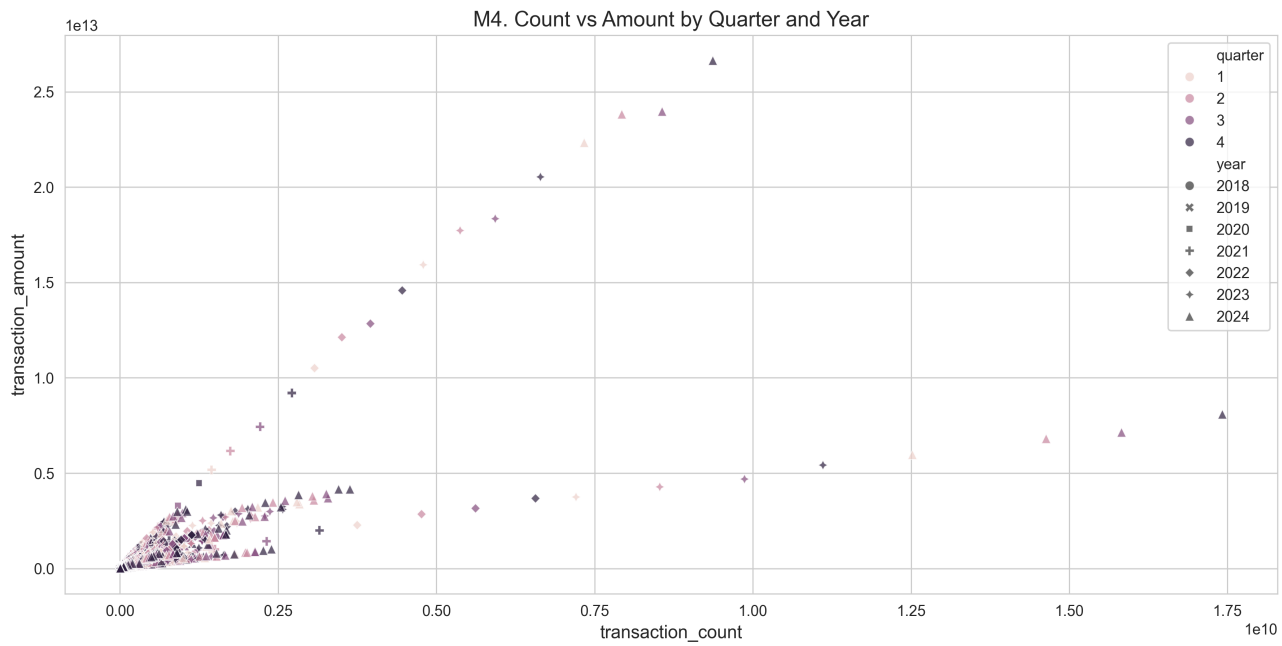

## M4. Count vs Amount by Quarter and Year

**Why this chart was chosen:** A segmented scatter plot is chosen to analyze relationship patterns while simultaneously incorporating time dimensions.

**Results:** Scatter plot created with transaction count and amount, segmented by quarter and year.

**Insights:** This chart shows whether the relationship between volume and value changes by quarter or year. It can reveal seasonal clustering or operational shifts in transaction behavior.

**Result:** Segmented scatter analysis reveals whether time periods alter the value-volume relationship.



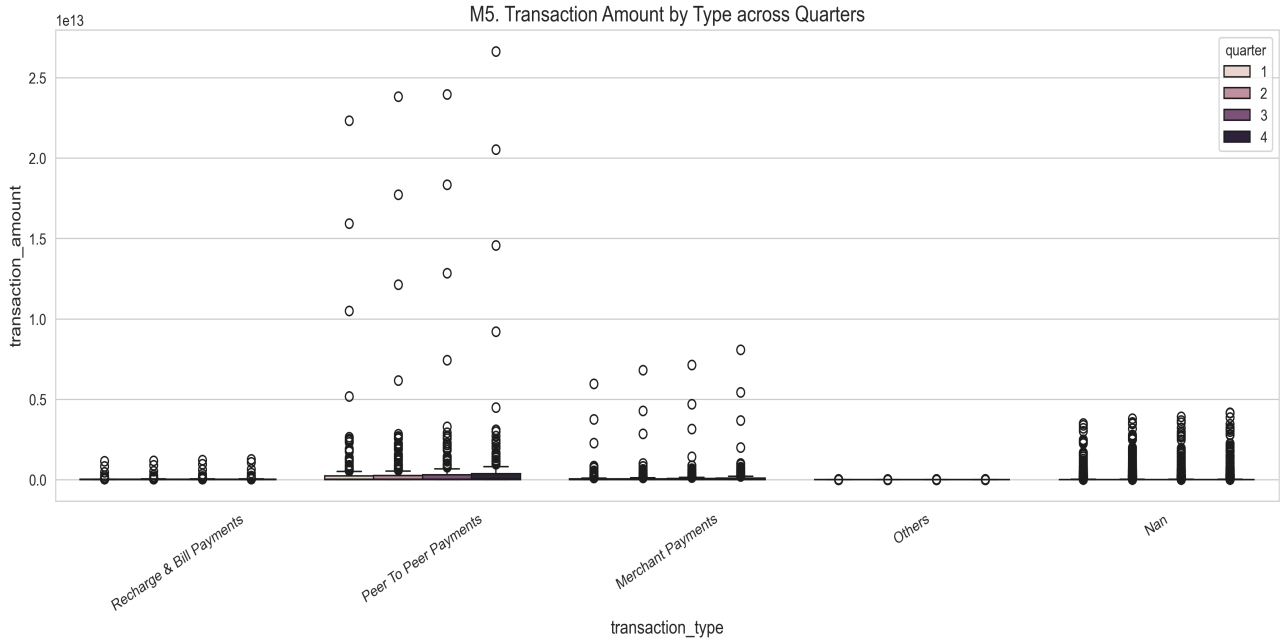

## M5. Transaction Amount by Type across Quarters

**Why this chart was chosen:** A grouped boxplot is chosen to compare spread, median, and outliers across categories and quarters together.

**Results:** Boxplot prepared for top transaction types across quarter segments.

**Insights:** This compares spread and variability within each transaction type across quarters. It helps identify high-variance categories, seasonal inconsistency, and stable performers.

**Result:** This view highlights category stability, seasonal fluctuation, and outlier-heavy payment segments.



# 4. Hypothesis Testing

To validate whether observed visual patterns are statistically meaningful, three hypothesis tests were conducted. These tests examine quarter-wise spending variation, performance difference between top and other states, and association between transaction type and quarter.

## H1. ANOVA: Average Transaction Amount across Quarters

**Why this was chosen:** ANOVA is chosen because quarter has more than two groups and we want to test whether average transaction amount differs significantly across them.

**Null Hypothesis (H0):** The mean transaction amount is the same across all quarters.

**Alternate Hypothesis (H1):** At least one quarter has a different mean transaction amount.

**Test Statistic:** 1.685422226979713

**P-Value:** 0.1677

**Result:** Fail to Reject H0

**Insights:** If H0 is rejected, at least one quarter has a statistically different average transaction amount. This supports seasonal spending variation and can guide quarter-specific campaigns.

## H2. T-Test: Top States vs Other States

**Why this was chosen:** An independent t-test is chosen to compare average transaction amount between two groups: top-performing states and remaining states.

**Null Hypothesis (H0):** There is no significant difference in mean transaction amount between top states and other states.

**Alternate Hypothesis (H1):** There is a significant difference in mean transaction amount between top states and other states.

**Test Statistic:** 17.26205150173197

**P-Value:** < 0.0001

**Result:** Reject H0

**Insights:** If rejected, leading states differ significantly from the rest in average transaction amount, confirming market concentration and the need for segmented regional strategy.

## H3. Chi-Square: Transaction Type vs Quarter

**Why this was chosen:** A chi-square test is chosen because both variables are categorical, and we want to test whether payment category distribution depends on quarter.

**Null Hypothesis (H0):** Transaction type and quarter are independent.

**Alternate Hypothesis (H1):** Transaction type and quarter are associated.

**Test Statistic:** 9.215790375755418

**P-Value:** 0.9545

**Result:** Fail to Reject H0

**Insights:** If rejected, transaction category distribution changes significantly by quarter. This supports seasonal or campaign-linked category shifts and targeted marketing by time period.

## 5. Hypothesis Test Summary Table

| Test Name | P-Value | Result |
|---|---|---|
| H1. ANOVA: Average Transaction Amount across Quarters | 0.1677 | Fail to Reject H0 |
| H2. T-Test: Top States vs Other States | < 0.0001 | Reject H0 |
| H3. Chi-Square: Transaction Type vs Quarter | 0.9545 | Fail to Reject H0 |

## 6. Business Insights

**1.** High-performing states account for a disproportionately large share of total transaction value, indicating strong regional concentration.

**2.** Transaction growth over time reflects increasing platform adoption and digital payment dependence.

**3.** Some payment categories dominate total transaction amount, suggesting areas where customer preference is strongest.

**4.** Quarter-wise variation in average ticket size suggests seasonal or campaign-linked user behavior.

**5.** The relationship between transaction count and amount indicates whether growth is volume-driven or value-driven.

**6.** Regional and category-level differences suggest the need for localized product, marketing, and retention strategies.

## 7. Client Recommendations

**1.** Prioritize top-performing states for premium campaigns, loyalty programs, and product cross-sell opportunities.

**2.** Develop region-specific marketing strategies because state and district performance is not uniform.

**3.** Invest more in the most valuable payment categories and analyze why they outperform others.

**4.** Use quarter-level trends to schedule campaigns during high-spending periods and improve performance during weaker quarters.

**5.** Monitor outlier regions and transaction spikes separately for fraud checks, campaign evaluation, and business opportunity discovery.

**6.** Use dashboard-driven real-time tracking to continuously monitor transaction amount, count, category, and geography.

## 8. Five Major Conclusions

**1.** PhonePe transaction activity is not evenly distributed; a limited number of states and categories drive a significant portion of the total value.

**2.** Transaction amount and transaction count are strongly connected, showing that higher usage generally leads to higher monetary throughput.

**3.** Quarter-wise and year-wise patterns indicate that digital payment activity changes over time and should be studied seasonally.

**4.** Category behavior differs across states, meaning regional user preference plays an important role in platform strategy.

**5.** Statistical testing supports that some observed differences are not random, strengthening the reliability of the analytical conclusions.

## 9. Final Project Summary

This project successfully transforms raw PhonePe transaction data into structured analytical intelligence. Through EDA, statistical testing, and visual reporting, it highlights growth trends, market concentration, regional disparities, seasonal patterns, and category contribution. The report supports both technical understanding and business decision-making, making it suitable for academic submission, portfolio presentation, and dashboard extension.